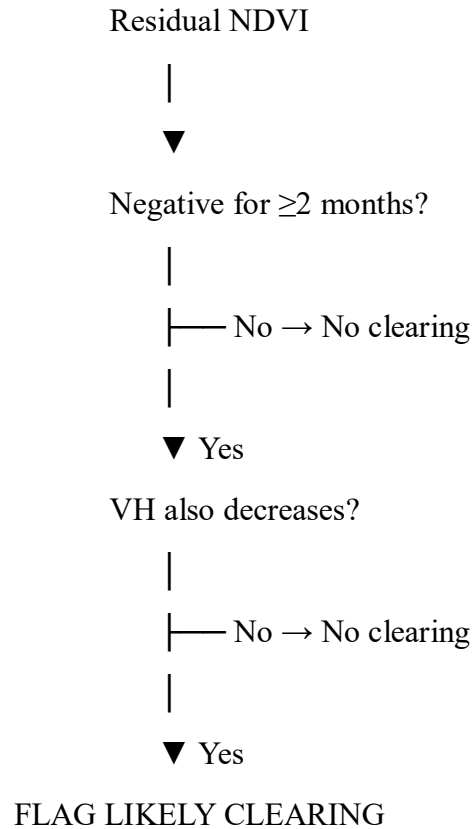


TASK 5 – REPORT

Detection rule:



1) Stable Forest Pixel:

Observation:

The rule remained quiet for most of the five-year record. However, it produced one apparent false positive during late 2020 (November–December), where consecutive negative residuals coincided with a temporary VH decrease despite no known clearing event. This likely reflects normal seasonal variability or radar noise rather than actual forest loss.

The pixels where false positive was triggered:

Residual_NDVI	VH	date	month	year
-0.016199291	-14.5176	2020-11	11	2020
-0.03713851	-16.4415	2020-12	12	2020

* Although the rule mathematically produces a false positive during late 2020, the NDVI residuals are extremely close to zero, suggesting negligible vegetation change. This indicates that the detected event is unlikely to represent a real clearing.

2) Dense Forest 1 Pixel: [LOSS YEAR: 2022]

Observation:

The residual NDVI series shows a clear and sustained negative anomaly beginning in May 2022 and continuing throughout the 2022 growing season, indicating a substantial reduction in vegetation relative to the healthy forest baseline. Unlike the raw NDVI series, the residuals remove seasonal variability, making the disturbance much easier to identify.

The combined detection rule successfully identifies the clearing during 2022 because the residual NDVI remains negative for multiple consecutive months and VH exhibits decreases during parts of the same period. However, VH does not decline continuously instead, it alternates between increases and decreases.

*The rule triggers during the clearing year (2022) but the VH values seem to be more variable than the NDVI Residual values, hence it should be interpreted with caution.

3) Dense Forest 2 Pixel: [LOSS YEAR: 2020]

Observation:

The NDVI residual exhibits a strong and persistent negative anomaly beginning in March 2020, clearly indicating vegetation loss. However, during the initial months of the disturbance, Sentinel-1 VH does not decrease as expected. Instead, VH becomes progressively less negative between March and July 2020, suggesting increased radar backscatter immediately following the disturbance. Only from August onward does VH begin to decrease while the NDVI residual remains negative, allowing the combined detection rule to trigger.

*Therefore, the rule successfully flags this pixel, but only after a noticeable delay relative to the onset of the NDVI disturbance.

4) Medium Forest Pixel: [LOSS YEAR: 2022]

Observation:

The Medium Forest pixel exhibits the normal seasonal residual pattern before 2022, with positive winter residuals and modest negative growing-season residuals. During the Hansen-reported loss year (2022), NDVI residuals remain negative for five consecutive months (May - September), satisfying the persistence requirement of the combined detection rule. Sentinel-1 VH backscatter also decreases during the early part of this period (particularly from May to July), allowing the rule to flag a potential disturbance.

However, both the NDVI anomaly and VH decline are relatively modest compared with the dense forest pixels, and VH continues to fluctuate throughout the time series. Consequently, although the combined rule is triggered, the evidence for a strong and sustained clearing event is weaker than in the dense forest cases.

* The combined rule would flag this pixel, but with considerably lower confidence as both the NDVI residuals and VH changes are comparatively small and less distinct.

5) Sparse Forest Pixel: [LOSS YEAR: 2020]

Observation:

The Sparse Forest pixel exhibits consecutive negative NDVI residuals during the 2020 growing season, satisfying the persistence component of the detection rule. However, Sentinel-1 VH backscatter does not show a consistent concurrent decline. Instead, VH alternates between increases and decreases throughout the same period, indicating normal temporal variability rather than a sustained structural change in the canopy.

* The Sparse Forest pixel satisfies the mathematical requirements of the combined rule because the NDVI residual remains negative for multiple consecutive months and VH decreases during part of the same interval (July-August 2020). However, the VH decline is relatively small and is embedded within frequent month-to-month fluctuations. Consequently, although the rule is technically triggered, the evidence for a genuine clearing event is weak.

6) Very Sparse Forest Pixel: [LOSS YEAR: 2020]

Observation:

The Very Sparse Forest pixel exhibits persistent negative NDVI residuals during the 2020 growing season, indicating vegetation conditions below the expected monthly climatology. However, similar negative residual patterns are already present before the reported loss year and continue throughout 2021–2023, suggesting long-term variability rather than a distinct disturbance centered on 2020.

Sentinel-1 VH backscatter does not show a consistent concurrent decline during the 2020 disturbance window. Instead, VH fluctuates between increases and decreases from month to month, without the sustained downward trend required by the combined detection rule.

*The Very Sparse Forest pixel satisfies the combined rule during part of the 2020 growing season because multiple consecutive negative NDVI residuals coincide with a short period of declining VH backscatter. However, VH exhibits substantial month-to-month variability before and after the reported loss year, and similar fluctuations occur elsewhere in the record. This reduces confidence that the detected signal represents a true canopy-clearing event.

Summary Table:

Pixel	Loss Year	Rule Triggered?	Correct Detection?	Confidence	Comments
Stable Forest	No loss	Yes (False Positive)	✗ No	Very Low	Trigger occurred only because of very small negative residuals and temporary VH variation. No real clearing exists.
Dense Forest 1	2022	Yes	✓ Yes	Moderate–High	Sustained NDVI anomaly with supporting VH decreases. Clear detection of disturbance.
Dense Forest 2	2020	Yes (Delayed)	✓ Yes	Moderate–High	NDVI detected clearing immediately, but VH initially behaved opposite to expectation. Rule triggered only later in the disturbance period.
Medium Forest	2022	Yes	✓ Yes	Moderate	Persistent NDVI anomaly and modest VH decrease. Clearing detected, although both signals are weaker than dense forest.
Sparse Forest	2020	Yes	⚠ Partially	Low	Rule technically triggered, but VH fluctuations were inconsistent. Evidence for real clearing is relatively weak.
Very Sparse Forest	2020	Yes	⚠ Partially	Low	Rule triggered mathematically, but similar NDVI and VH patterns occur throughout the record, reducing confidence in the detection.

GEE Code Link: <https://code.earthengine.google.com/3fda0c65f24745ac37fbffb3e623b8f2>

GitHub Repository Link: <https://github.com/SiddharthBhimpure/bavaria-forest-change-analysis.git>

END OF REPORT